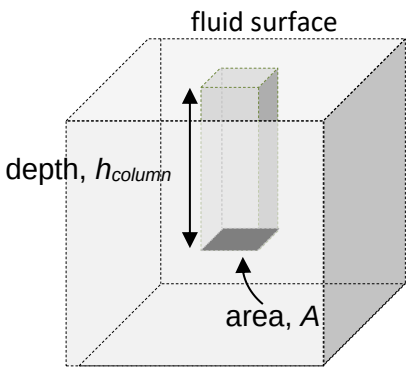
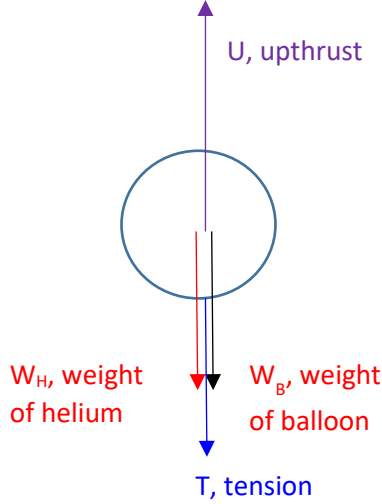


1	(a)	Consider a column of fluid as shown. Pressure of fluid acting on the column of fluid at a depth h_{column}, W_{column} is the weight of column of fluid, m_{column} is the mass of column of fluid, A is the base area of the column of fluid $p_{\text{fluid}} = \frac{W_{\text{column}}}{A} = \frac{m_{\text{column}}g}{A} \quad \Leftarrow p = \frac{F}{A}$ $= \frac{V_{\text{column}}\rho_{\text{fluid}}g}{A} \quad \Leftarrow \rho = \frac{m}{V}$ $= h_{\text{column}}\rho_{\text{fluid}}g$ [B1] Correct application of pressure and density. [B1] Correctly define A, W_{column}, m_{column} and the steps.	
	(b)	(i)	The pressure due to a fluid increases with depth in the fluid. [B1] For the balloon in air, there is a difference in air pressure over the upper and lower surface of the balloon, which gives rise to upthrust on the balloon. [B1]
		(ii)	

			upthrust, $U = V\rho g$ $= \frac{4}{3}\pi(0.15)^3(1.2)(9.81) \quad [\text{C1}]$ $= 0.166 \text{ N}$ At equilibrium $T + W_H + W_B = U$ $T = 0.166 - \left(\frac{2.9}{1000}\right)(9.81) - \frac{4}{3}\pi(0.15)^3(0.18)(9.81) \quad [\text{C1}]$ $= 0.113 \text{ N} \quad [\text{A1}]$
		(iii)	Helium diffuses through tiny pores in the balloon/ helium leaks out of the balloon. [B1]

2	(a)	$\Delta KE = KE - KE$
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